

Alberto Rosas

AI/ML/Software Engineer

PROFESSIONAL SUMMARY

AI/ML Engineer with a decade of software engineering experience, specializing in production AI/ML systems. Successfully transitioned from traditional software development to AI engineering, bringing strong fundamentals in system architecture and production deployment. Experienced in building multi-agent systems, implementing RAG architectures, and deploying LLMs for business intelligence applications.

TECHNICAL SKILLS

Machine Learning & AI

- Production experience with GPT-4, Claude, Gemini, xAI and open-source models (Llama 3, Mistral)
- Multi-agent orchestration using LangChain/LangGraph
- RAG implementation with vector databases (Qdrant, MongoDB Atlas, Chroma, FAISS...)
- Fine-tuning, prompt and context engineering for domain-specific applications
- PyTorch, Transformers, scikit-learn for model development
- ML FTI architecture approach

Data Engineering & Processing

- ETL pipeline design for structured/unstructured data
- Database optimization and schema design for ML workflows
- Experience with SQL/NoSQL databases (PostgreSQL, MySQL, MongoDB, Redis)
- Data validation, cleaning, and feature engineering
- MCP (Model Context Protocol) server implementation

Software Engineering & DevOps

- Python, TypeScript/JavaScript, PHP

- Microservices architecture, RESTful APIs, event-driven systems
 - Docker, Kubernetes, AWS (EC2, S3, RDS, Lambda)
 - CI/CD with GitHub Actions/Gitlab, comprehensive testing practices
 - MLOps: MLflow, Weights & Biases, DVC for experiment tracking and model versioning
 - LLMOps: Langfuse, Opik, Langsmith for LLM-specific monitoring and evaluation
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PROFESSIONAL EXPERIENCE

AI Engineer | Storage360 (*Contract*)

July 2024 – Present

Developing SADIE (Storage360 AI Data Intelligence Engine), a natural language interface for business analytics that enables non-technical users to query complex operational data.

- Architected multi-agent system using LangGraph for orchestrating query understanding, SQL generation, and result synthesis workflows
- Built data preprocessing pipeline handling diverse business schemas, reducing query response time to sub-300ms through intelligent caching
- Implemented MCP servers for secure database connectivity, enabling real-time access to distributed data sources
- Developed comprehensive evaluation framework tracking query accuracy, latency, and user satisfaction metrics, as well as RAG-related evaluation metrics like hallucination, context precision and recall.
- Processed and prepared training data from historical queries for model fine-tuning and validation.

Technical Lead / Senior Software Engineer | IncFile (*Consultant*)

December 2022 – Present

Leading technical initiatives while hands-on building AI-powered internal tools for business process automation.

- Designed multi-agent application for document processing and decision support, handling 1,000+ daily requests
- Implemented memory systems (short-term, long-term, procedural) for maintaining context across user sessions
- Built Hybrid RAG pipeline integrating business documentation with vector search for intelligent query responses as well as constant improvement through research and experimentation.

- Reduced manual data entry by 65% through intelligent form processing and validation systems
- Established engineering standards for AI development including testing frameworks and deployment practices
- Mentored team of 8 engineers in modern ML practices and production deployment strategies

Senior Software Engineer | Global Cybersec

May 2017 – June 2020

Developed security automation platforms and incident response systems for enterprise clients.

- Built integrated security platform connecting SIEM, IDS/IPS, and firewall systems for automated threat response
- Designed event-driven architecture processing millions of daily security events for pattern recognition
- Reduced incident response time by 60% through automated correlation and alerting systems
- Developed data pipelines for log aggregation and analysis across 100+ security devices
- Created dashboards and reporting tools for security metrics and regulatory requirements

Senior Full-Stack Developer (*Multiple Companies*)

June 2020 – December 2022

Progressive roles at NetActuate, Fulcrum Digital, Online MedEd, HQrental Software

- Contributed to healthcare SaaS platform serving 10,000+ medical professionals
- Implemented microservices architecture using RabbitMQ for event-driven communication
- Built and maintained RESTful APIs with comprehensive test coverage (85%+)
- Deployed applications on AWS infrastructure using Docker and Kubernetes
- Developed real-time data synchronization systems for distributed applications

KEY PROJECTS & ACHIEVEMENTS

Natural Language to SQL System: Built production system enabling business users to query databases using natural language, achieving 86% accuracy on complex multi-table joins

GraphRAG Implementation: Developed hierarchical knowledge graph for improving context retrieval in RAG systems, reducing hallucination rate by 40%

ADDITIONAL INFORMATION

- **Languages:** English (Native), Spanish (Native)
- **Writing:** Technical blog on AI/ML implementations and system design
- **Current Focus:** Exploring advanced RAG techniques and multi-modal AI applications